

Fabrication and Training on Solar-Powered Mobile Phone Charger with Torch: A Community Education Project in Laduk VDC, Nepal

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Abstract—This paper presents the design, fabrication, and community-based training of a low-cost solar-powered mobile phone charger integrated with an LED torch, carried out under the Community Education Project (CEP) of the Department of Electrical and Electronics Engineering, Kathmandu University. The activity was conducted in Laduk Village Development Committee (VDC), Dolakha district, Nepal, an off-grid mountainous area with limited access to electricity and technical education. A team of five undergraduate students, under faculty supervision, organized a four-day outreach program that combined household-level need assessment with hands-on technical training for secondary-level students at two local schools. Students were introduced to basic principles of electricity and safe handling practices before being trained to assemble a solar-powered charger unit using locally available and economical components. The paper discusses the motivation, planning, fabrication approach, and pedagogical methodology used to deliver the training, along with the outcomes and broader implications of the project for rural electrification and technical skill development. The project demonstrates that small-scale, student-led interventions combining device fabrication with hands-on training can help bridge the technical education gap in remote communities.

Index Terms—solar charger, rural electrification, community education, off-grid energy, technical training, renewable energy, Nepal

I. INTRODUCTION

Access to reliable electricity remains a significant challenge in many remote and mountainous regions of Nepal, where grid infrastructure is either unavailable or unreliable. In such areas, solar photovoltaic (PV) based standalone devices offer a practical and low-cost means of meeting basic energy needs such as mobile phone charging and lighting. Beyond the provision of hardware, however, the sustainability of such interventions depends heavily on local technical capacity – the ability of community members to understand, maintain, and eventually replicate such systems.

The Community Education Project (CEP) is an undergraduate outreach initiative of the Department of Electrical and Electronics Engineering at Kathmandu University, through

which student teams design and deliver small-scale technical interventions in underserved communities. This paper documents one such initiative, titled “Fabrication and Training on Solar Mobile Phone Charger and Solar Torch,” carried out by a team of five students under faculty supervision in Laduk VDC, Dolakha district. The program combined a household- and school-level assessment of the local demographic, social, academic, infrastructural, and economic conditions with hands-on fabrication training delivered to secondary school students.

The remainder of this paper is organized as follows. Section II states the objectives and rationale of the project. Section III describes the field methodology, including site selection, logistics, and program execution. Section IV presents the design and fabrication approach for the solar charger–torch unit. Section V describes the training methodology used with students. Section VI discusses outcomes, impact, and challenges encountered. Section VII concludes the paper.

II. OBJECTIVES AND RATIONALE

The project was motivated by a broader survey of the academic and infrastructural condition of Laduk VDC. The area was found to have a generally satisfactory academic environment, with two schools serving the local population; however, students seeking higher-level education were required to travel to Charikot or the Kathmandu Valley. The principal gap identified was the absence of a skilled, technically trained local workforce capable of performing day-to-day technical tasks and imparting technical knowledge to students.

Based on this assessment, the specific objectives of the project were defined as follows:

- To introduce secondary-level students to the fundamental principles of electricity and safe electrical practices.
- To train students to fabricate a functional, low-cost solar-powered mobile phone charger with an integrated LED torch using economically available components.

- To leave behind a working prototype device and transferable technical skill within the school community.
- To contribute, in a small way, toward building local technical capacity that could translate into livelihood opportunities such as household wiring and basic electronics repair (e.g., radios, televisions, and mobile phones).

III. FIELD METHODOLOGY

A. Site Selection and Approach

Laduk VDC, located in Dolakha district, was selected as the destination in coordination with the president of the local school committee. Prior to the visit, communication was established with the school administration to confirm the schedule and scope of the training program.

B. Logistics

Reaching Laduk required an overland journey to Singati followed by an approximately 3.5-hour trek across mountainous terrain. The team arrived at the destination late in the evening after navigating difficult trail conditions and rainfall. Local arrangements for lodging were coordinated by the school committee. On subsequent days, an additional walk of approximately one hour was required to reach the school from the place of stay.

C. Program Execution

The four-day program followed a structured schedule:

- 1) **Day 1:** Travel to Laduk VDC and arrival at the accommodation.
- 2) **Day 2:** A coordination meeting with school teachers to plan the training session, followed by classroom delivery at Shanti Adarsha Higher Secondary School for students of grades eight, nine, and ten.
- 3) **Day 3:** A repeat training session at Chitre Secondary School for students of grades seven, eight, and nine.
- 4) **Day 4:** Closing formalities with the school president, principals, students, and local hosts, followed by departure to Kathmandu.

At each school, the session began with an introduction of the team and the motivation behind the visit, followed by an explanation of basic electrical concepts and safety practices. The team then divided responsibilities: a subset of members led the conceptual explanation while the remaining members guided students through the hands-on fabrication process, including instruction on the use of a soldering iron and an overview of the electronic components involved. Interactive and engaging activities were interspersed throughout the sessions to sustain student attention and participation. At the conclusion of the training at each site, a completed prototype of the solar-powered charger with torch was handed over to the school.

IV. DESIGN AND FABRICATION OF THE SOLAR CHARGER–TORCH UNIT

The device fabricated and demonstrated during the training was designed to be simple, low-cost, and buildable with readily available components, so that the underlying skill could

be replicated by students and community members without specialized equipment.

A. System Overview

At a conceptual level, the unit consists of four functional blocks, illustrated in the following signal path:

- 1) **Solar PV panel** – converts incident sunlight into DC electrical energy.
- 2) **Charge control and protection stage** – a blocking diode and basic regulation circuitry to prevent battery back-discharge through the panel at night and to limit overcharging.
- 3) **Rechargeable battery** – a small-capacity cell (e.g., Li-ion or NiMH) that stores the harvested solar energy for use when sunlight is unavailable.
- 4) **Output stage** – a USB-compatible output for mobile phone charging and an LED driver circuit providing torch functionality, with a switch to select between charging and lighting modes.

B. Design Considerations

The design emphasized affordability and ease of assembly over performance optimization, since the primary purpose of the exercise was pedagogical. Component selection favored parts that are inexpensive, robust to basic handling errors by first-time learners, and available through common local electronics suppliers. Screw terminals and simple soldered joints were used in place of printed circuit boards to keep the assembly process accessible to students with no prior soldering experience.

C. Hands-on Assembly

Students were guided step by step through the assembly process: identifying and testing components, understanding polarity, soldering connections, and assembling the enclosure. This hands-on approach was intended not merely to produce a working device but to build procedural familiarity with basic electronics fabrication that students could apply to future repair or construction tasks.

V. TRAINING METHODOLOGY

The pedagogical approach combined conceptual instruction with immediate practical application, following a learn-by-doing model suited to a one-day classroom session:

- **Conceptual grounding:** A brief, accessible explanation of electricity, current, and safety precautions was provided before any hands-on work, ensuring that students understood the basic principles behind the device they were about to build.
- **Guided fabrication:** Students assembled the solar charger–torch unit under direct supervision, with team members circulating to assist individually.
- **Engagement activities:** To maintain enthusiasm and focus among younger students, short interactive and entertaining activities were incorporated between technical segments.

- **Tangible takeaway:** A completed working prototype was left with each school, reinforcing the practical value of the session and providing a reference device for future use.

This approach was repeated with minor adaptation at both Shanti Adarsha Higher Secondary School and Chitre Secondary School, allowing the team to refine delivery between the first and second sessions.

VI. OUTCOMES, IMPACT, AND CHALLENGES

A. Outcomes

The project delivered technical training to students across grades seven through ten at two schools in Laduk VDC and left behind functional solar charger–torch prototypes at each institution. Students gained first-hand exposure to basic electrical concepts, component identification, and soldering – skills not typically covered in the standard curriculum available locally.

B. Broader Impact

The project reflects a scalable model in which technical education is delivered alongside a tangible, useful device, rather than through abstract instruction alone. As noted in the project’s motivation, the acquisition of basic technical skills – such as household wiring or the repair of common electronic equipment (radios, televisions, and mobile phones) – can translate into local livelihood opportunities in communities where skilled technical labor is scarce. Even a small-scale intervention of this kind can serve as a catalyst for broader technical awareness within a community.

C. Challenges

The project also surfaced practical challenges common to remote outreach work, including difficult terrain and long trekking distances to reach both the accommodation and the schools, adverse weather conditions during transit, and the logistical constraint of delivering a complete conceptual-to-practical training session within a single classroom visit per school.

VII. CONCLUSION

This paper presented a student-led Community Education Project that combined the fabrication of a low-cost solar-powered mobile phone charger with an integrated torch and its delivery as a hands-on technical training exercise for secondary school students in Laduk VDC, Dolakha district, Nepal. By pairing a needs assessment of the local academic and infrastructural context with practical, replicable fabrication training, the project sought to address a specific gap: the shortage of local technical skill and knowledge in a remote mountain community. The experience demonstrates that compact, well-planned outreach initiatives – even when constrained to a few days and a small team – can meaningfully introduce renewable energy concepts and practical technical skills to underserved communities, with the potential to support future livelihood opportunities through basic electrical and electronics competence.

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